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M.Sc. (IT) Integrated College & B.Sc. Data Science College

Affiliated To Veer Narmad South Gujarat University



ONLINE WATCHES WEBSITE / WATCHSTORE

A Project Report Submitted in Partial Fulfillment of the Requirements

for the Degree of

Bachelor of Computer Applications (B.C.A.) Semester – V

Subject: 504 Web Framework & Services (WFS)

Academic Year: 2024 – 2025

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Group No.: 129

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1 INTRODUCTION

Online Watches Website (WatchStore) is a web-based e-commerce application developed as part of the BCA Semester 5 course in Web Framework & Services (WFS). The main goal of this project is to provide a clean and user-friendly online shopping platform where customers can browse authentic wristwatches from top brands such as Titan, Fossil, Citizen, Seiko, Samsung, Casio, and Timex.

Traditional watch shopping requires customers to physically visit multiple retail stores to compare designs, specifications, prices, and brand warranties. This project solves this problem by allowing users to explore different watch categories such as Luxury Watches, Chronograph Watches, Smart Watches, Casual & Minimalist, and Formal & Classic from the comfort of their home or mobile device.

The system is divided into two primary modules:

- **Client / Customer Side:** Allows users to register an account, securely log in, browse watches, filter by brand or category, search for specific models, save favorites in their wishlist, manage items in a shopping cart, proceed to checkout, make payments using Razorpay Test Mode, leave product reviews with star ratings, and submit general feedback.
- **Admin Side:** Provides store managers with a protected administrative portal to view store metrics, manage watch inventory (add, edit, delete products), manage watch brands with logos, and moderate customer reviews.

1.1 Project Profile

Project Title	Online Watches Website / WatchStore
Course / Subject	BCA Semester 5 – 504 Web Framework & Services (WFS)
Academic Year	2024 – 2025
Developed By	Shubham Ashish Pancholi (Roll No. 3382) Saumya Chaudhari (Roll No. 3235) Vrushank Changawala (Roll No. 3234)
Project Guide	Minal Desai
Group Number	129
Front-End Technologies	HTML5, CSS3, JavaScript
Server-Side Language	PHP 8.x (Core PHP)
Database Management	MySQL / MariaDB (Database: watches_db)
Development Environment	XAMPP Server (Apache + MySQL), Visual Studio Code
Payment Gateway	Razorpay Standard Checkout (Test Mode)
Target Users	Online watch shoppers, watch enthusiasts, and store administrators

2 ENVIRONMENT DESCRIPTION

The environment description outlines the minimum hardware configurations and software packages required to develop, host, and execute the Online Watches Website.

2.1 Hardware & Software Requirements

2.1.1 Hardware Requirements

Development / Server Machine:

- Processor: Intel Core i3 / i5 or AMD Ryzen equivalent (2.0 GHz or higher)
- RAM: 4 GB minimum (8 GB recommended for smooth XAMPP and browser execution)
- Hard Disk Space: 500 MB free storage for project files, images, and database
- Input Devices: Standard Keyboard and Mouse / Touchpad
- Display: 1366 x 768 resolution or higher

Client / End-User Machine:

- Device: Desktop, Laptop, Tablet, or Smartphone
- RAM: 1 GB minimum
- Network: Active Internet or Local Area Network (LAN) connection

2.1.2 Software Requirements

Server Environment:

- Operating System: Windows 10 / Windows 11 (64-bit)
- Web Server: Apache HTTP Server 2.4 (included in XAMPP)
- Database Server: MySQL 5.7+ / MariaDB 10.4+
- Server-Side Scripting: PHP 8.1 / 8.2 with cURL and OpenSSL extensions enabled

Client Environment:

- Web Browser: Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari
- JavaScript: Enabled in browser for client-side interactivity and Razorpay checkout

2.2 Technology Used

The project is built using standard web technologies that are part of the BCA curriculum:

- **PHP (Core PHP 8.x):** Handles server-side business logic, session management, user authentication with `password_hash()` and `password_verify()`, SQL query preparation, and API communication with Razorpay via cURL.
- **MySQL:** Stores all persistent data including users, admins, watch products, brands, orders, order items, wishlist records, reviews, and feedback messages.
- **HTML5:** Provides clean and semantic markup for forms, navigation menus, product cards, tables, and modal dialogs.

- **CSS3:** Custom styling for modern aesthetics, responsive grid layouts, card designs, buttons, and badges without heavy third-party framework dependencies.
- **JavaScript (Vanilla JS):** Handles dynamic client-side interactions such as quantity changes, image switching, and launching the official Razorpay Standard Checkout popup.
- **Razorpay Test Mode:** Integrated using official Razorpay cURL APIs for server-side order generation and client-side checkout popup for testing online payments without real money.
- **XAMPP:** Provides an all-in-one local development stack containing Apache, MySQL, and PHP.

3 SYSTEM REQUIREMENT SPECIFICATION

3.1 Data Dictionary

The database for the project is named `watches_db` and contains 9 relational tables. Below is the complete data dictionary detailing the field names, data types, constraints, and descriptions for all tables.

Table 1: admins

Stores credentials for administrators who manage the store catalog and reviews.

Field Name	Data Type	Constraint	Description
admin_id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for administrator
username	VARCHAR(50)	NOT NULL, UNIQUE	Unique admin login username
email	VARCHAR(100)	NOT NULL, UNIQUE	Admin email address
password	VARCHAR(255)	NOT NULL	Bcrypt hashed admin password
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date and time admin was registered

Table 2: users

Stores customer profile and authentication information.

Field Name	Data Type	Constraint	Description
user_id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique customer identifier
full_name	VARCHAR(100)	NOT NULL	Customer full name
email	VARCHAR(100)	NOT NULL, UNIQUE	Customer email used for login
mobile	VARCHAR(15)	NOT NULL	10-digit mobile contact number
password	VARCHAR(255)	NOT NULL	Encrypted password using Bcrypt
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account registration timestamp

Table 3: products

Stores the catalog of watches offered in the store.

Field Name	Data Type	Constraint	Description
id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique watch product ID
name	VARCHAR(150)	NOT NULL	Watch title / model name
brand	VARCHAR(100)	NOT NULL	Brand name (e.g. Titan, Fossil)
price	DECIMAL(10,2)	NOT NULL	Price of watch in INR (₹)
image	VARCHAR(255)	NOT NULL	Relative file path to product image
category	VARCHAR(100)	NOT NULL	Category (Luxury, Smart, etc.)

description	TEXT	NOT NULL	Detailed specifications & description
quantity	INT(11)	NOT NULL, DEFAULT 10	Available stock inventory count
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Product creation timestamp

Table 4: brands

Stores watch manufacturer brand names and logos.

Field Name	Data Type	Constraint	Description
id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique brand identifier
name	VARCHAR(100)	NOT NULL, UNIQUE	Brand name
logo	VARCHAR(255)	NOT NULL	Path to brand logo image
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Brand record creation timestamp

Table 5: orders

Stores customer orders, delivery addresses, and payment transaction details.

Field Name	Data Type	Constraint	Description
id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Internal order ID
order_number	VARCHAR(30)	NOT NULL, UNIQUE	Generated order reference number
user_id	INT(11)	NULL	ID of user who placed order
customer_name	VARCHAR(100)	NOT NULL	Name of recipient
customer_email	VARCHAR(100)	NOT NULL	Email for order confirmation
customer_phone	VARCHAR(15)	NOT NULL	Phone number for delivery contact
shipping_address	TEXT	NOT NULL	Detailed street address
city	VARCHAR(50)	NOT NULL	Delivery city
pincode	VARCHAR(10)	NOT NULL	Postal PIN code
payment_method	VARCHAR(50)	NOT NULL, DEFAULT 'Razorpay'	Method of payment
payment_status	VARCHAR(20)	NOT NULL, DEFAULT 'Pending'	Status: Pending / Paid
total_amount	DECIMAL(10,2)	NOT NULL	Total order value in INR
order_status	VARCHAR(20)	NOT NULL, DEFAULT 'Placed'	Status: Pending Payment / Placed
razorpay_order_id	VARCHAR(100)	NULL	Razorpay order reference ID
razorpay_payment_id	VARCHAR(100)	NULL	Razorpay transaction ID
razorpay_signature	VARCHAR(255)	NULL	HMAC SHA256 verified signature
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Order placement timestamp

Table 6: order_items

Stores individual line items associated with each order.

Field Name	Data Type	Constraint	Description
id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique order item identifier
order_id	INT(11)	FOREIGN KEY (orders.id)	Associated parent order ID
product_id	INT(11)	NOT NULL	ID of ordered watch product
product_name	VARCHAR(150)	NOT NULL	Name of watch at time of purchase
price	DECIMAL(10,2)	NOT NULL	Unit price of watch
quantity	INT(11)	NOT NULL, DEFAULT 1	Purchased quantity
subtotal	DECIMAL(10,2)	NOT NULL	Line item total (price * quantity)

Table 7: wishlist

Stores watches bookmarked by customers for future purchase.

Field Name	Data Type	Constraint	Description
id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique wishlist record ID
user_id	INT(11)	NULL	Customer user ID
product_id	INT(11)	FOREIGN KEY (products.id)	Watch product ID
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date item added to wishlist

Table 8: reviews

Stores product-specific customer reviews and 1 to 5 star ratings.

Field Name	Data Type	Constraint	Description
id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique review identifier
product_id	INT(11)	FOREIGN KEY (products.id)	ID of watch reviewed
user_id	INT(11)	FOREIGN KEY (users.user_id)	ID of user who wrote review
rating	INT(1)	NOT NULL	Customer rating (1 to 5 stars)
comment	TEXT	NOT NULL	Written review feedback text
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Review submission timestamp

Table 9: feedback

Stores general website and service feedback submitted by visitors.

Field Name	Data Type	Constraint	Description
id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique feedback ID
name	VARCHAR(100)	NOT NULL	Visitor / customer name
email	VARCHAR(100)	NOT NULL	Contact email address

rating	INT(1)	NOT NULL, DEFAULT 5	Website rating (1 to 5)
message	TEXT	NOT NULL	Feedback comment message
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Feedback submission timestamp

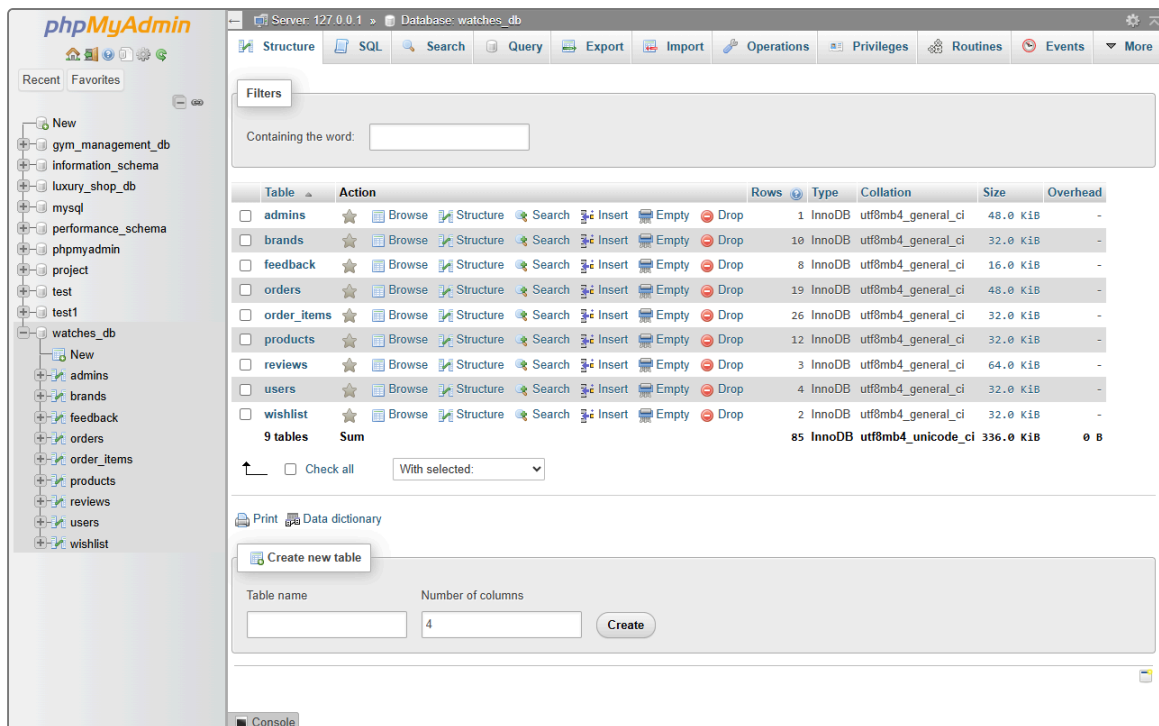


Figure 3.1: Actual phpMyAdmin View of watches_db Database Tables

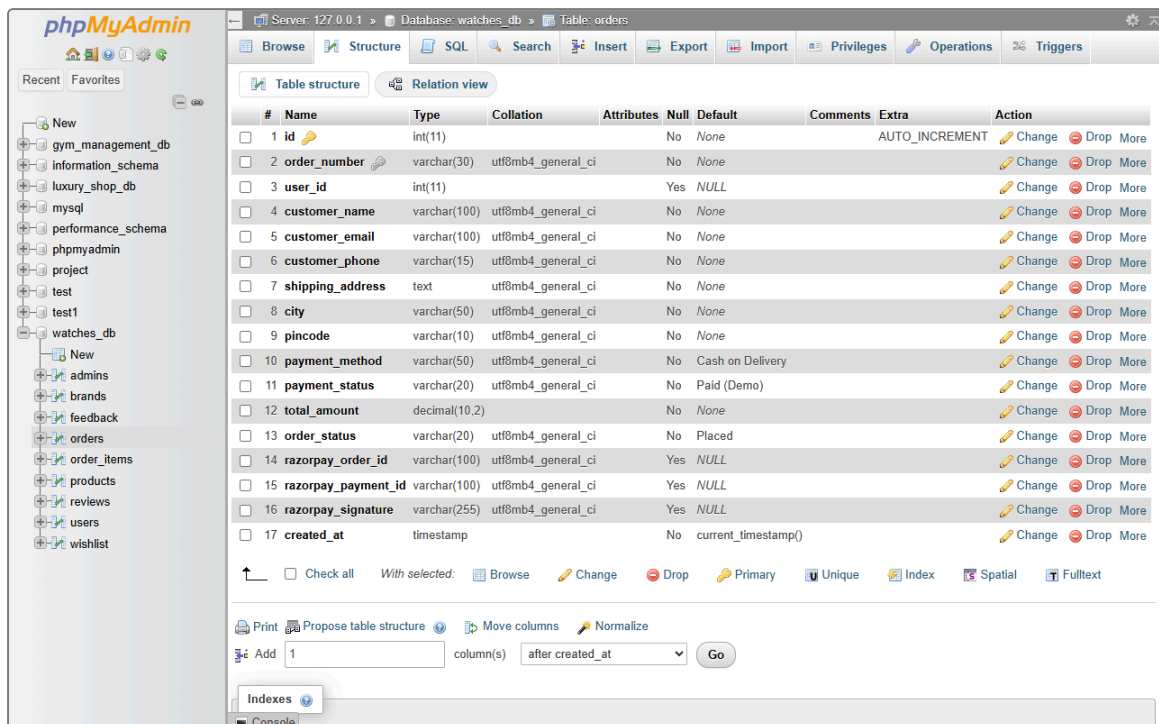


Figure 3.2: Actual MySQL Table Structure for orders Table

3.2 Feasibility Analysis

Before developing the software, a feasibility study was conducted to evaluate whether the proposed online watch store is viable from economic, technical, and operational perspectives.

3.2.1 Economic Feasibility

The project is highly feasible economically because it utilizes open-source technologies that do not incur any licensing or software acquisition costs.

- PHP, MySQL, and Apache HTTP Server are free and open-source under GPL/Apache licenses.
- XAMPP is freely available for Windows local hosting.
- Visual Studio Code is a free code editor.
- Razorpay Test Mode provides free sandbox API credentials for testing transactions without setup fees or payment deductions.
- Existing personal computers and laptops were used for development, eliminating new hardware expenses.

3.2.2 Technical Feasibility

The technical requirements for this project are modest and fully supported by standard computer hardware available in college labs and homes.

- Core PHP 8 provides robust support for session handling, database connectivity via `mysqli`, and `cURL` API calls.
- MySQL handles relational data integrity with primary and foreign keys.
- Standard web browsers support HTML5, CSS3, and JavaScript out of the box without requiring third-party plugins.
- The team possessed the required foundational knowledge of PHP, SQL queries, and web design taught in the BCA curriculum.

3.2.3 Behavioural Feasibility

The system is designed with an intuitive interface that requires no prior training:

- Customers are already familiar with standard online shopping patterns: browsing items, clicking "Add to Cart", and checking out.
- The checkout page displays clear input fields with instant validation messages.
- The admin portal has a clean navigation menu allowing administrators to easily add or edit products without technical expertise.

3.3 E-R Diagram

The Entity-Relationship (E-R) diagram represents the conceptual database structure of the Online Watches Website. It illustrates the primary entities ('USERS', 'PRODUCTS', 'BRANDS', 'ORDERS', 'ORDER_ITEMS', 'WISHLIST', 'REVIEWS', 'FEEDBACK', 'ADMINS'), their key attributes, and the relational cardinality connecting them.

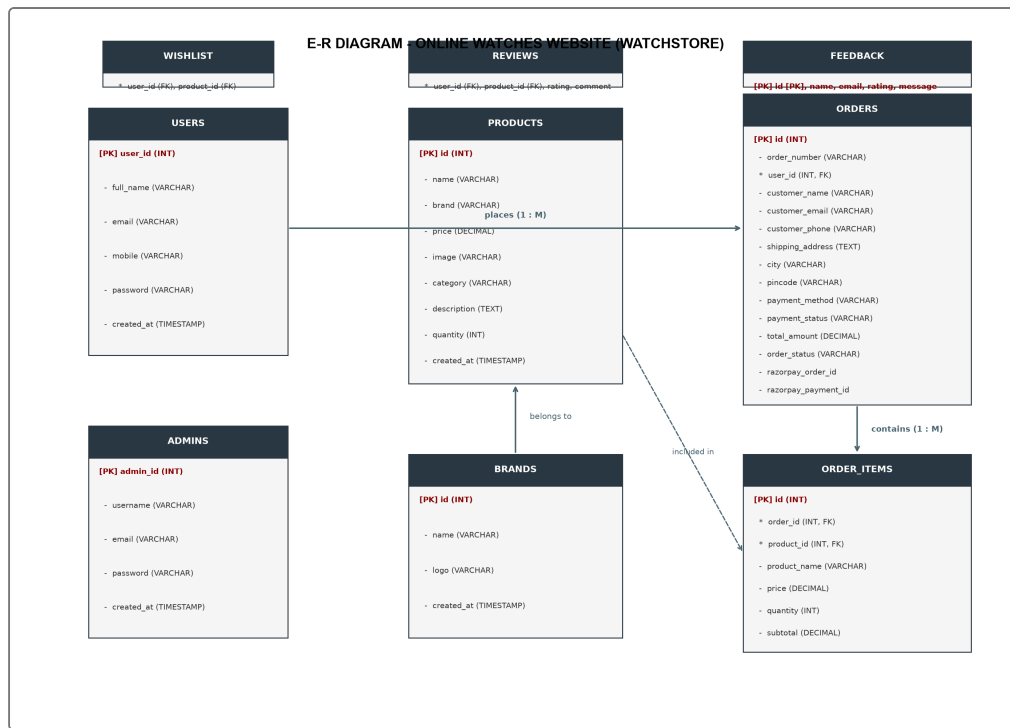


Figure 3.3: E-R Diagram for Online Watches Website

Key Relationships Explained:

- **USERS to ORDERS (1 : M)**: A registered user can place multiple orders over time, but each order belongs to one specific customer.
- **ORDERS to ORDER_ITEMS (1 : M)**: Each order can contain one or more watch line items.
- **PRODUCTS to ORDER_ITEMS (1 : M)**: A watch product can appear in multiple customer order items.
- **BRANDS to PRODUCTS (1 : M)**: A watch brand (e.g. Titan, Fossil) manufactures multiple watch models in the catalog.
- **USERS to REVIEWS to PRODUCTS (M : N via reviews)**: A user can review specific watches, and each watch can have reviews from multiple users.
- **USERS to WISHLIST to PRODUCTS (M : N via wishlist)**: A user can save multiple watches to their wishlist for future purchase.

3.4 UML Diagrams

Unified Modeling Language (UML) diagrams provide visual blueprints of the system's actors, structural classes, and operational workflows.

3.4.1 Use Case Diagram

The Use Case Diagram depicts the system boundary, the two primary actors ('Customer / Visitor' and 'Administrator'), and the actions they can perform.

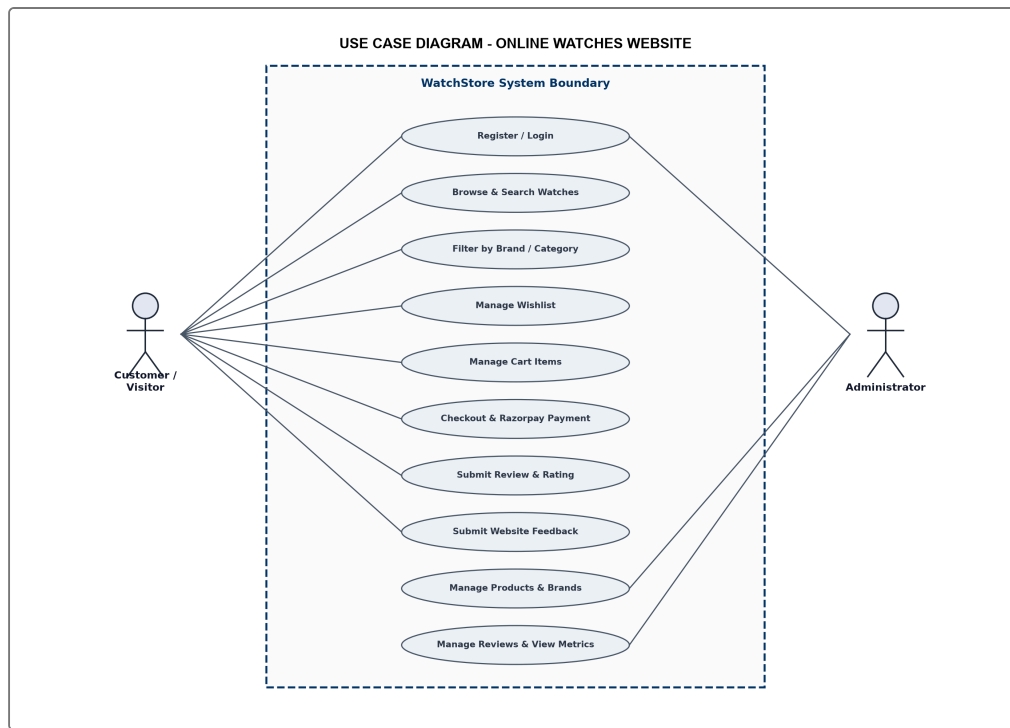


Figure 3.4: UML Use Case Diagram

3.4.2 Class Diagram

The Class Diagram models the object-oriented structure of the system, showing classes, attributes, methods, and relationships.

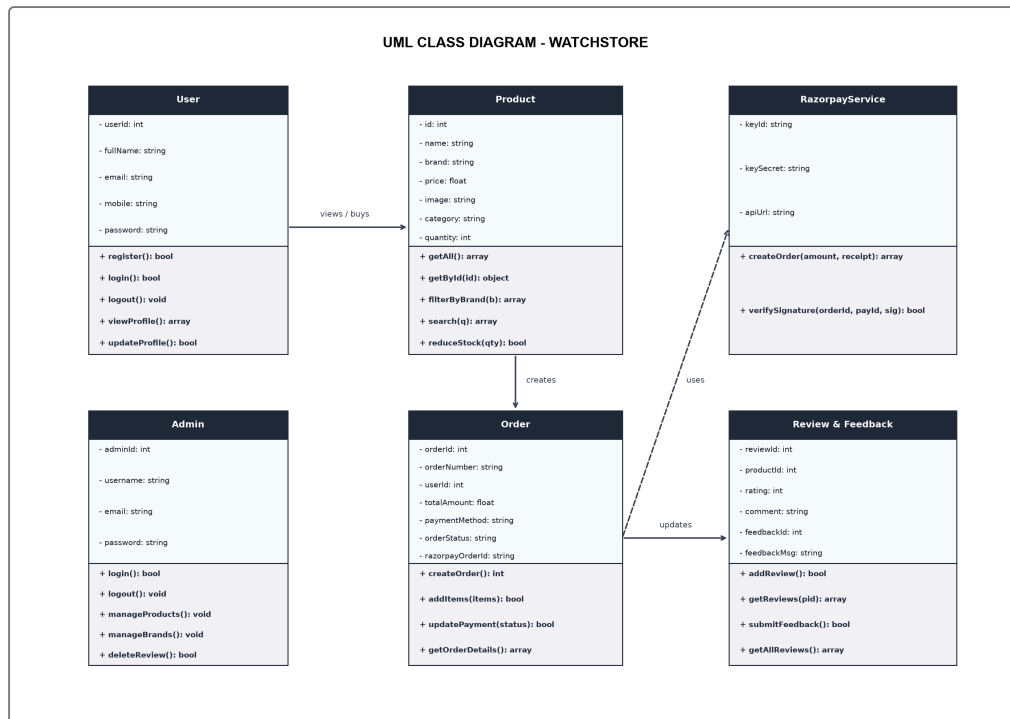


Figure 3.5: UML Class Diagram

3.4.3 Activity Diagram

The Activity Diagram illustrates the step-by-step business process flow for customer product browsing, cart management, checkout, and payment processing.

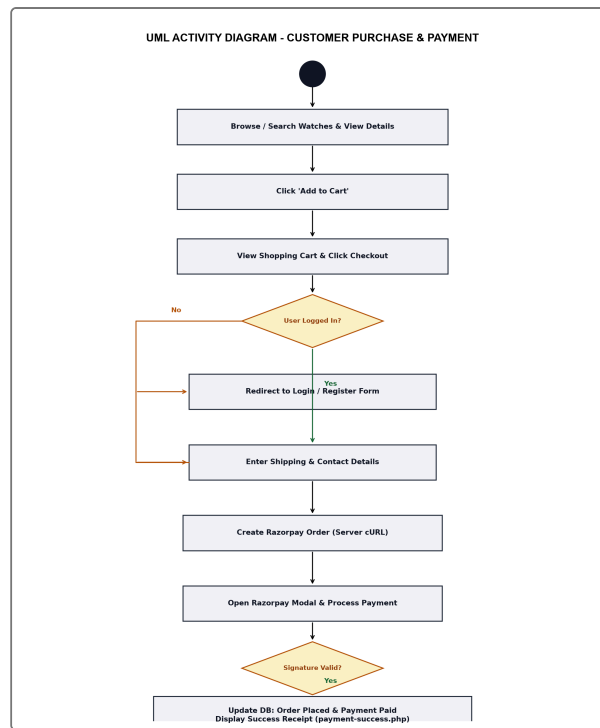


Figure 3.6: UML Activity Diagram for Customer Purchase & Payment

3.4.4 Sequence Diagram

The Sequence Diagram documents the chronological message flow between the Customer Browser, PHP Checkout Controller, MySQL Database, and Razorpay API during payment verification.

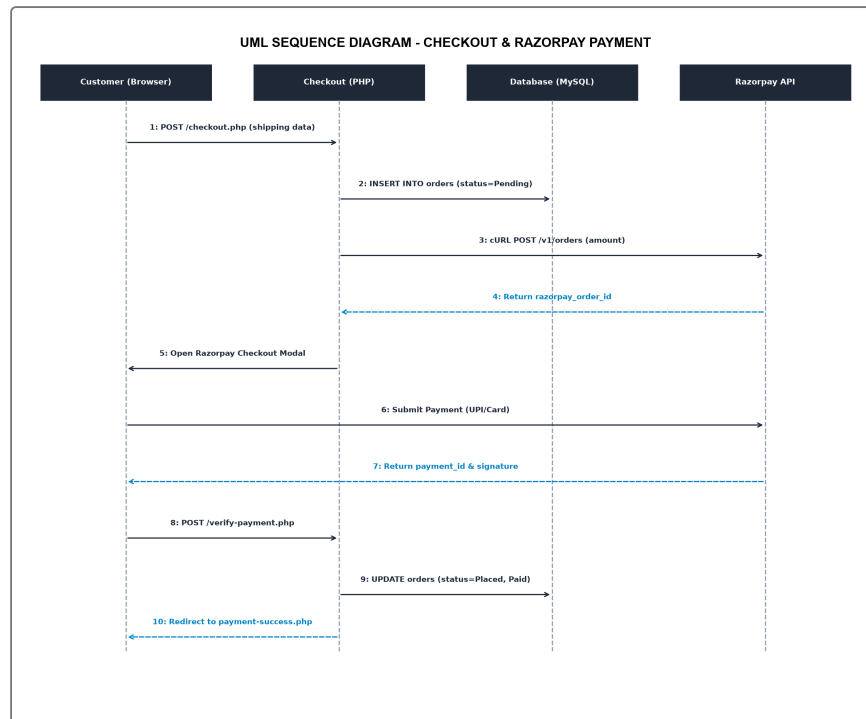


Figure 3.7: UML Sequence Diagram for Checkout & Razorpay Payment

4 SOFTWARE PROJECT PLANNING

4.1 Scope

The scope of the Online Watches Website project encompasses the design, development, and testing of a functional e-commerce prototype tailored for wristwatches:

- Developing a secure customer registration and login system with encrypted passwords.
- Creating an online watch catalog categorized into Luxury, Chronograph, Smart, Casual, and Formal watches.
- Enabling brand-based filtering for major brands such as Titan, Fossil, Citizen, Seiko, Samsung, ColorFit, Daniel Wellington, Timex, Casio, and Battuta.
- Implementing search functionality to quickly find watches by keywords.
- Providing an interactive shopping cart and wishlist system.
- Implementing a checkout form with shipping address collection and integration with Razorpay Test Mode for payment simulation.
- Enabling customer product reviews with 1 to 5 star ratings and a general website feedback form.
- Providing an administrative back-end to manage products, brands, and customer reviews.

4.2 Objectives

- To create an easy-to-use, responsive web interface for watch buyers.
- To replace manual in-store visits with 24/7 digital product browsing and order placement.
- To practice core web development concepts taught in BCA Semester 5 (Core PHP, MySQL relational design, session security).
- To understand how third-party payment gateways (Razorpay) integrate with PHP web applications using server-side cURL APIs.
- To maintain accurate inventory counts by automatically decrementing product stock when orders are successfully placed.

4.3 Expected Advantages

- **Convenience:** Users can browse specifications, high-quality images, and prices anytime without visiting physical showrooms.
- **Transparent Information:** Real-time pricing, stock availability, and genuine customer reviews help buyers make informed decisions.
- **Simulated Online Payment:** Demonstrates real-world e-commerce checkout flow using Razorpay Standard Checkout in test mode without financial risk.
- **Centralized Store Management:** Store administrators can easily update watch prices, add new arrivals, and manage brand partnerships through a single admin dashboard.

5 SYSTEM DESIGN

System design encompasses the input forms through which data enters the application and the output views through which information is presented to users and administrators.

5.1 Input Design

Input design ensures that data entered by users is validated for accuracy, completeness, and security before being processed or stored in the database.

5.1.1 User Registration Form

Collects customer full name, email address, 10-digit mobile number, and password. The system checks if the email is already registered and hashes passwords with Bcrypt before storing.

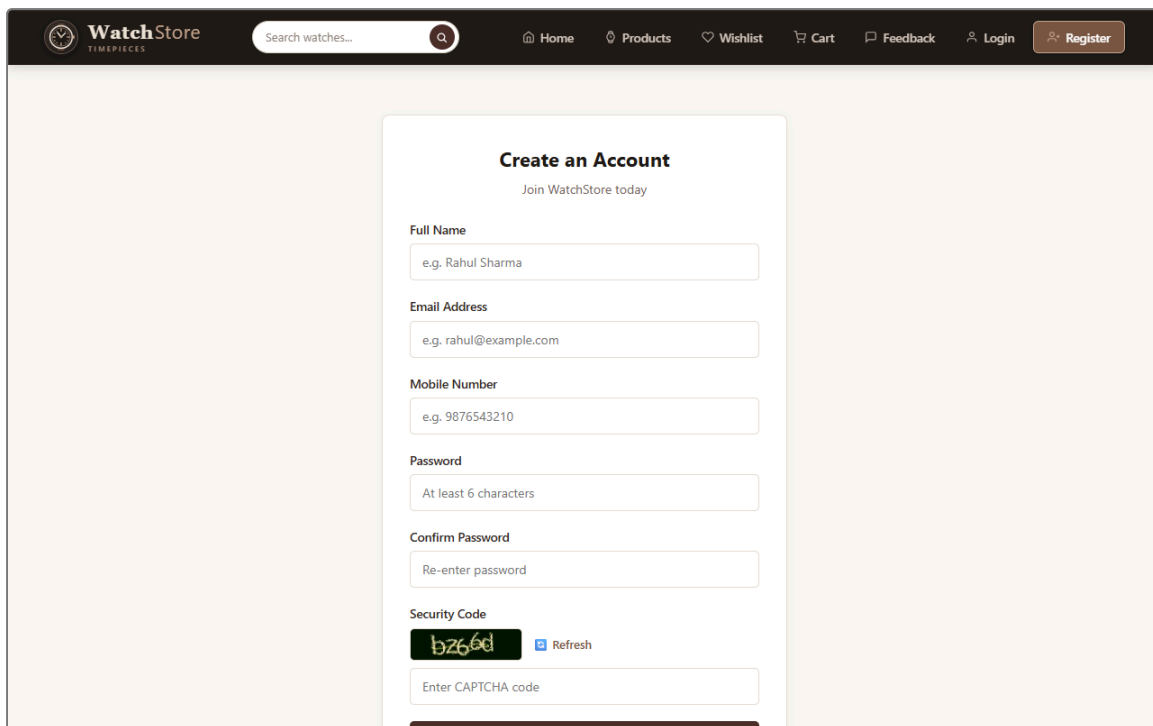


Figure 5.1: Customer Registration Page (register.php)

5.1.2 User Login Form

Authenticates returning customers using their registered email and password. Upon successful authentication, a PHP session is established.

WatchStore
TIMEPIECES

Search watches...

Home Products Wishlist Cart Feedback Login Register

User Login
Sign in to your WatchStore account

Email Address
e.g. user@example.com

Password
Enter your password

Security Code
 Refresh
 Enter CAPTCHA code

Login

Don't have an account? [Register here](#)

Home Products Wishlist Cart Feedback Login
© 2026 WatchStore. All rights reserved.

Figure 5.2: Customer Login Page (login.php)

5.1.3 Checkout & Delivery Information Form

Collects the recipient's name, email, phone number, complete shipping address, city, and PIN code. Allows selecting Razorpay as the payment method.

WatchStore
TIMEPIECES

Search watches...

Home Products Wishlist Cart Feedback Login Register

User Login
Sign in to your WatchStore account

Please log in to proceed with checkout and place your order.

Email Address
e.g. user@example.com

Password
Enter your password

Security Code
 Refresh
 Enter CAPTCHA code

Login

Don't have an account? [Register here](#)

Figure 5.3: Checkout & Delivery Form (checkout.php)

5.1.4 Product Review & Rating Form

Located on the product details page. Allows logged-in users to submit a 1 to 5 star rating and written comment for the specific watch.

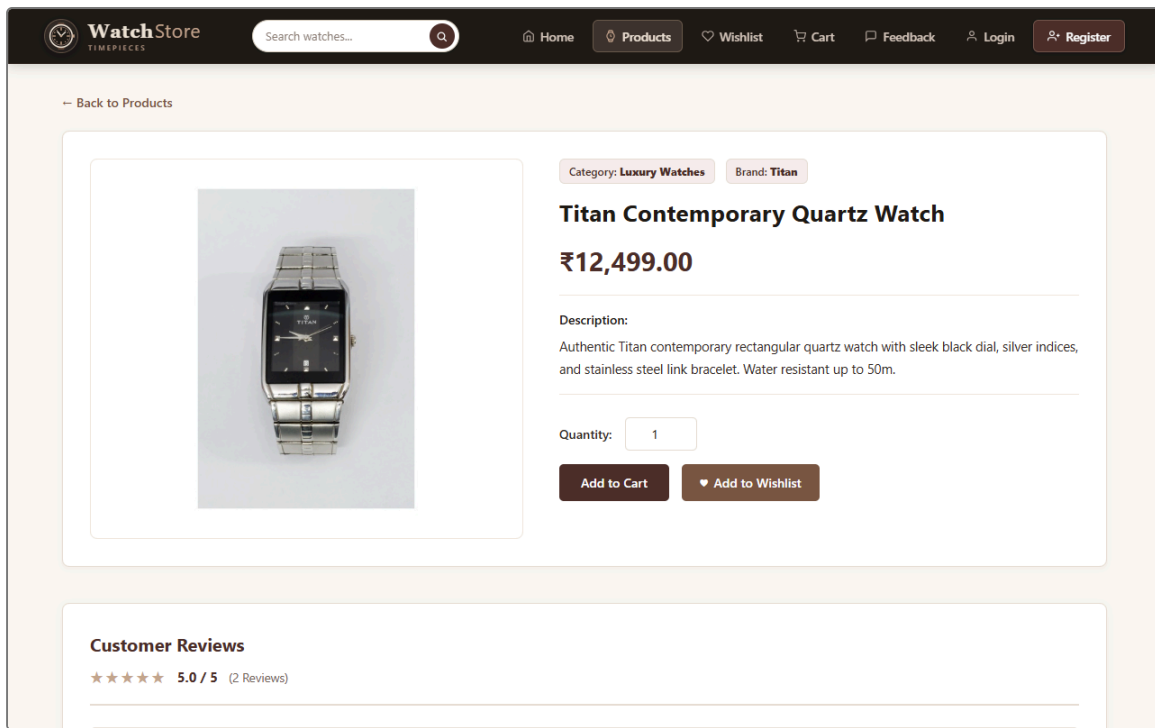


Figure 5.4: Product Details & Review Submission (product-details.php)

5.1.5 Website Feedback Form

Enables site visitors to submit general feedback about their shopping experience, website usability, and service ratings.

The screenshot shows the 'Customer Feedback' form. The form has three main sections: 'Your Name' with a text input field, 'Email Address' with a text input field, and 'Feedback / Message' with a larger text area. A 'Submit Feedback' button is located at the bottom of the form. The footer of the page includes navigation links (Home, Products, Wishlist, Cart, Feedback, Login) and a copyright notice: '© 2026 WatchStore. All rights reserved.'

Figure 5.5: Website Feedback Page (feedback.php)

5.1.6 Admin Login Form

Dedicated authentication gateway for store managers to access administrative controls.

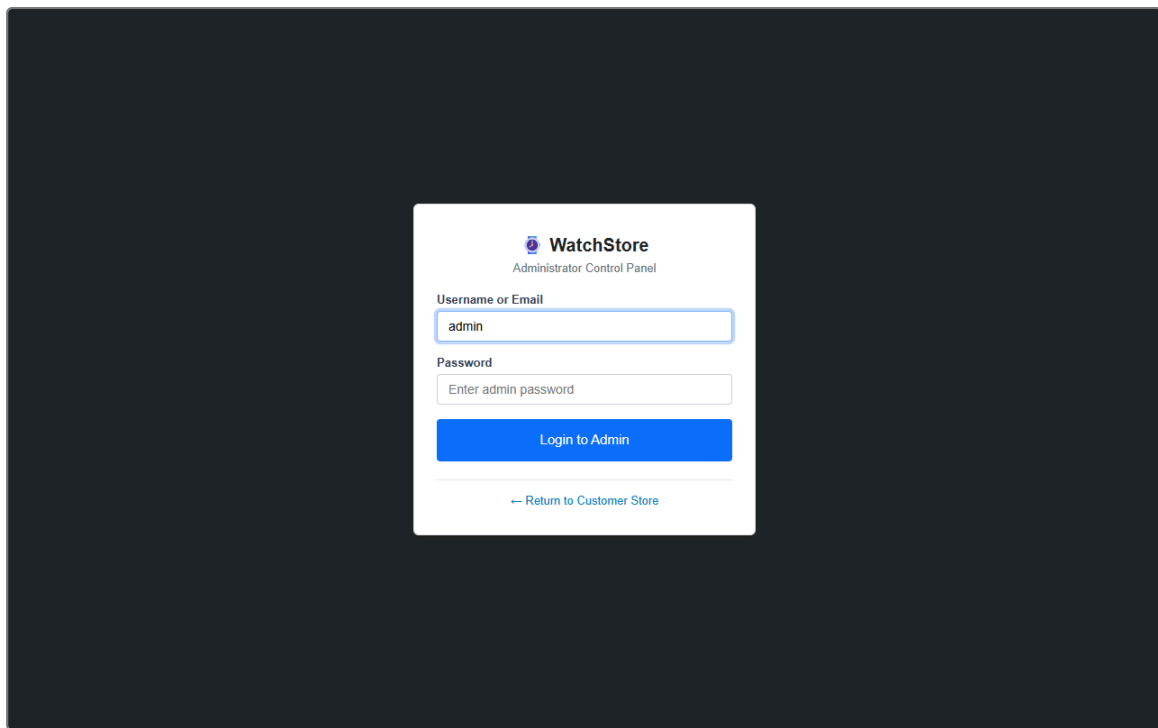


Figure 5.6: Admin Login Portal (admin/login.php)

5.1.7 Admin Add Product Form

Enables administrators to add new watches to the store by inputting model name, brand, price, category, stock quantity, description, and product image.

Figure 5.7: Admin Add Product Page (admin/product_add.php)

5.1.8 Admin Add Brand Form

Allows store administrators to add new watch brands and upload brand logos.

Figure 5.8: Admin Add Brand Page (admin/brand_add.php)

5.2 Output Design

Output design focuses on presenting information clearly and attractively to customers and administrators.

5.2.1 Home Page

The landing page features the store branding, hero banner, featured watch categories, brand logos carousel, and latest arrivals.

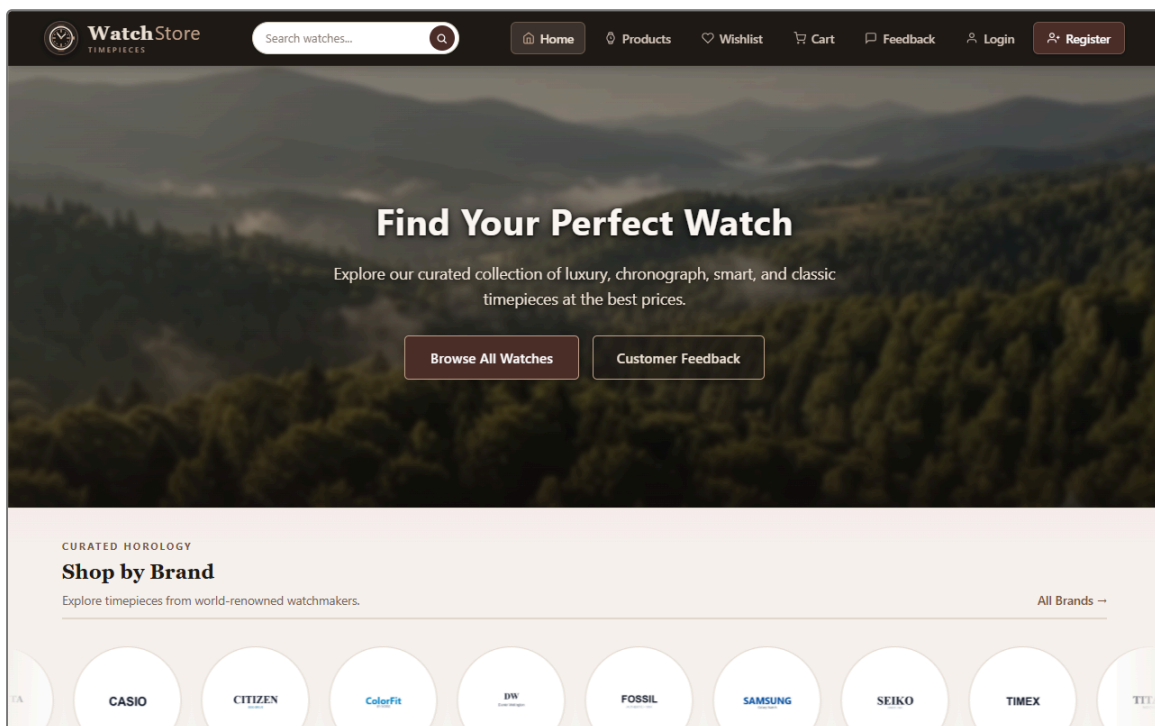


Figure 5.9: WatchStore Home Page (index.php)

5.2.2 Products Catalog Page

Displays all available watches in a responsive grid layout with product images, brand tags, prices in INR, and quick action buttons for Wishlist and Details.

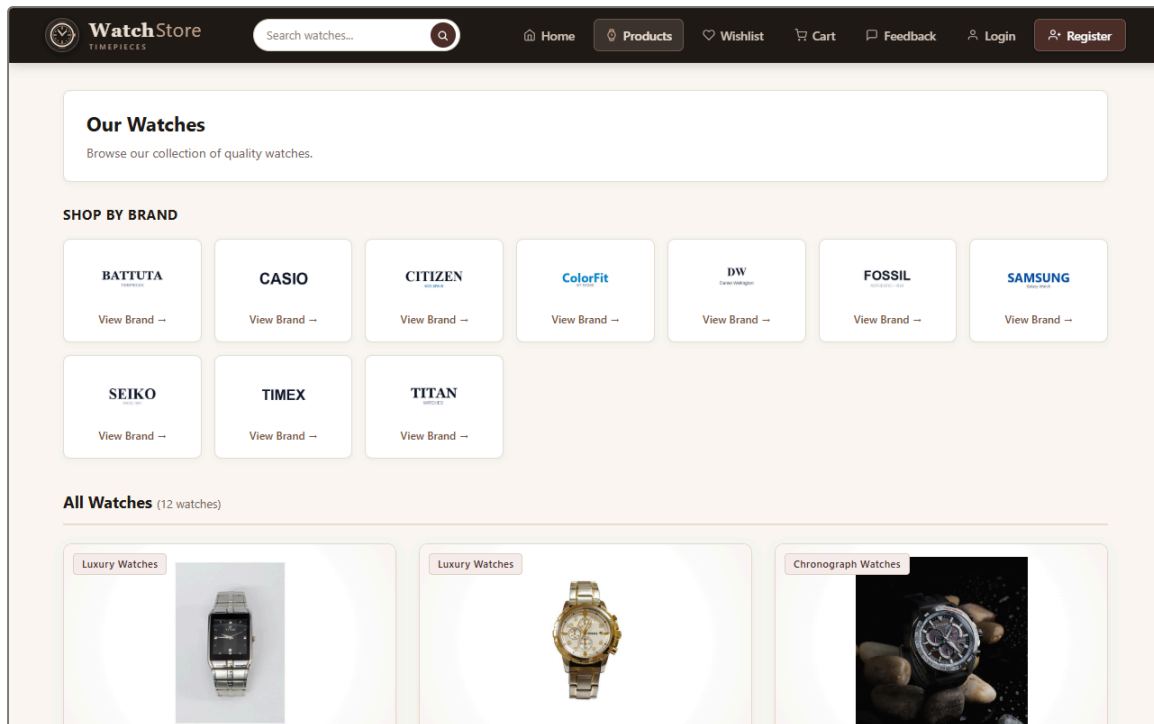


Figure 5.10: Watch Products Catalog (products.php)

5.2.3 Brand Filter Page

Filters the watch collection dynamically based on the selected brand (e.g. Titan, Fossil, Casio).

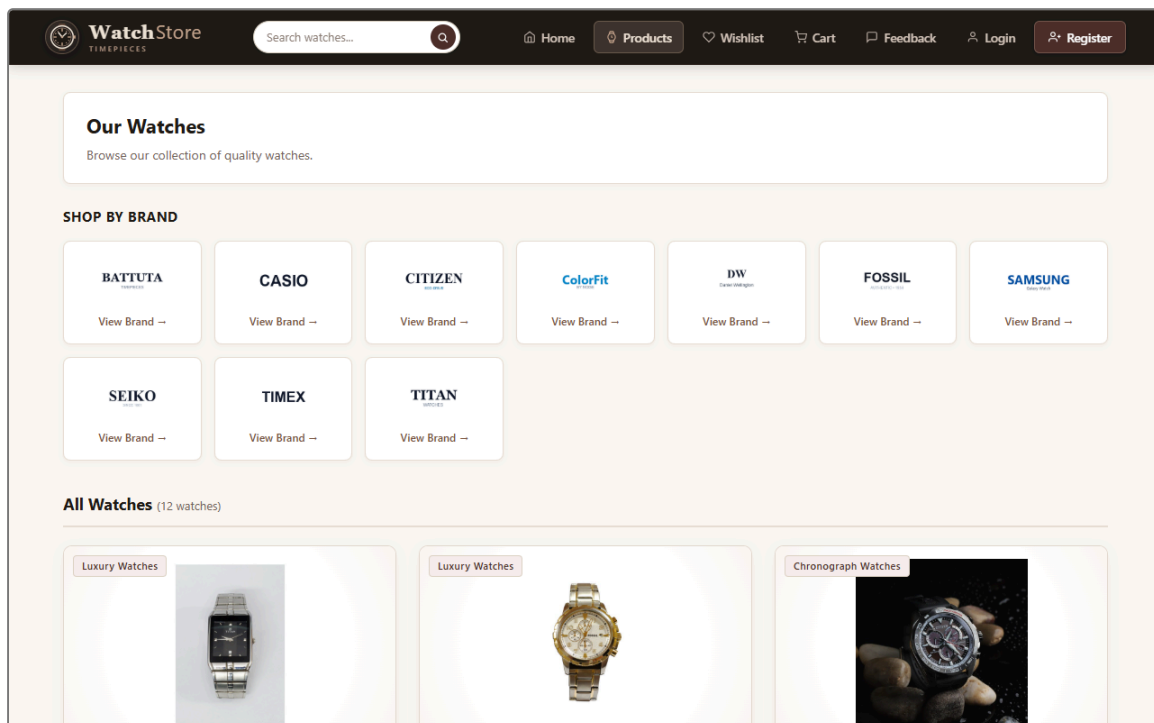


Figure 5.11: Brand Specific Watch Filter (brand.php)

5.2.4 Shopping Cart Page

Displays all items currently in the customer's cart, showing watch thumbnails, unit prices, quantity modifiers, subtotal, and grand total.

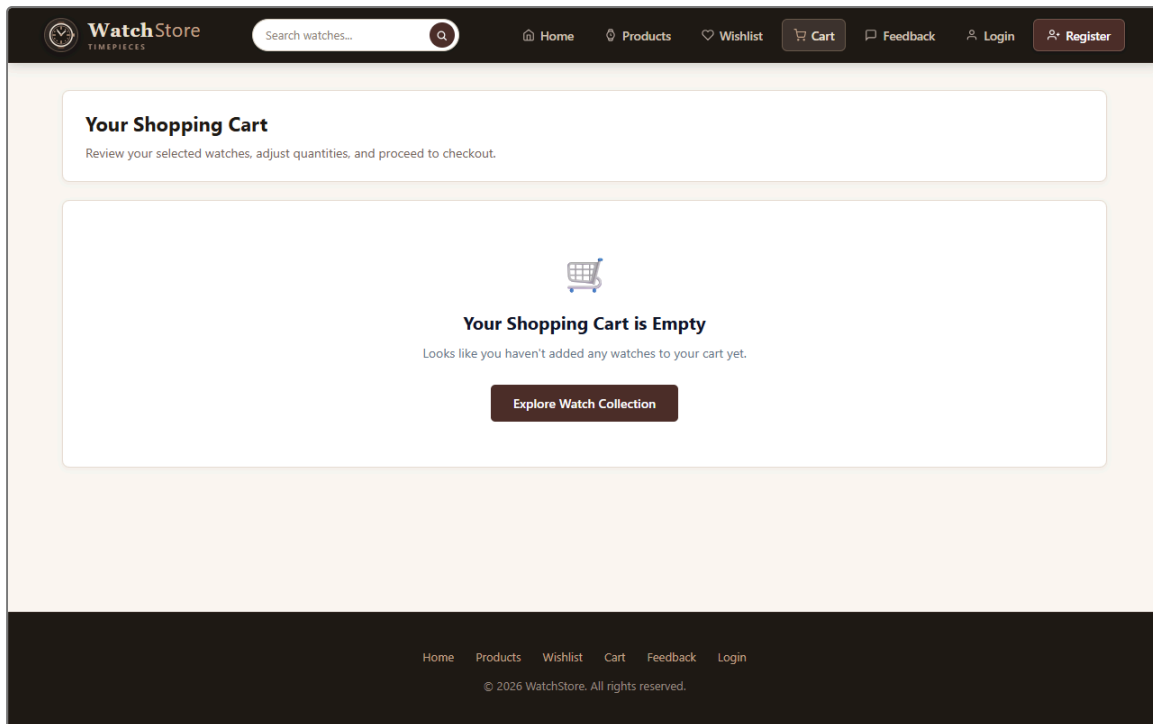


Figure 5.12: Shopping Cart Page (cart.php)

5.2.5 Wishlist Page

Displays watches saved by the customer for future reference with options to quickly transfer them to the cart or remove them.

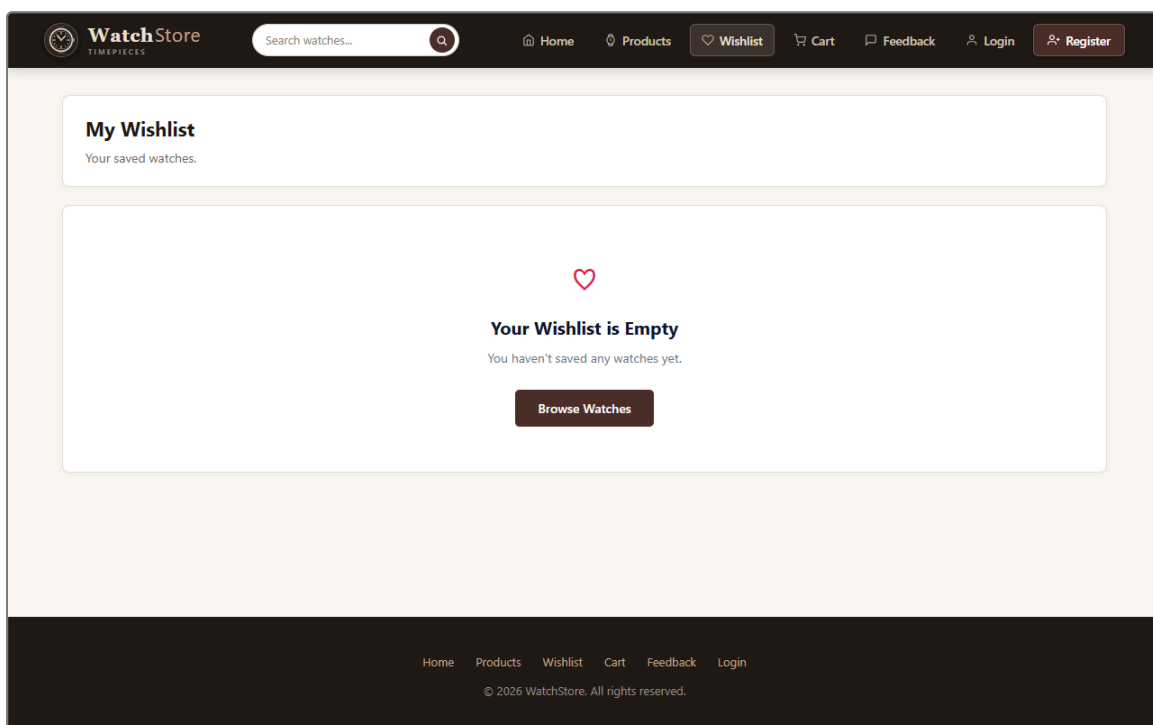


Figure 5.13: Customer Wishlist Page (wishlist.php)

5.2.6 Order Confirmation Receipt

Generated after successful payment verification. Displays the unique order reference number, Razorpay Payment ID, shipping address, and ordered item breakdown.

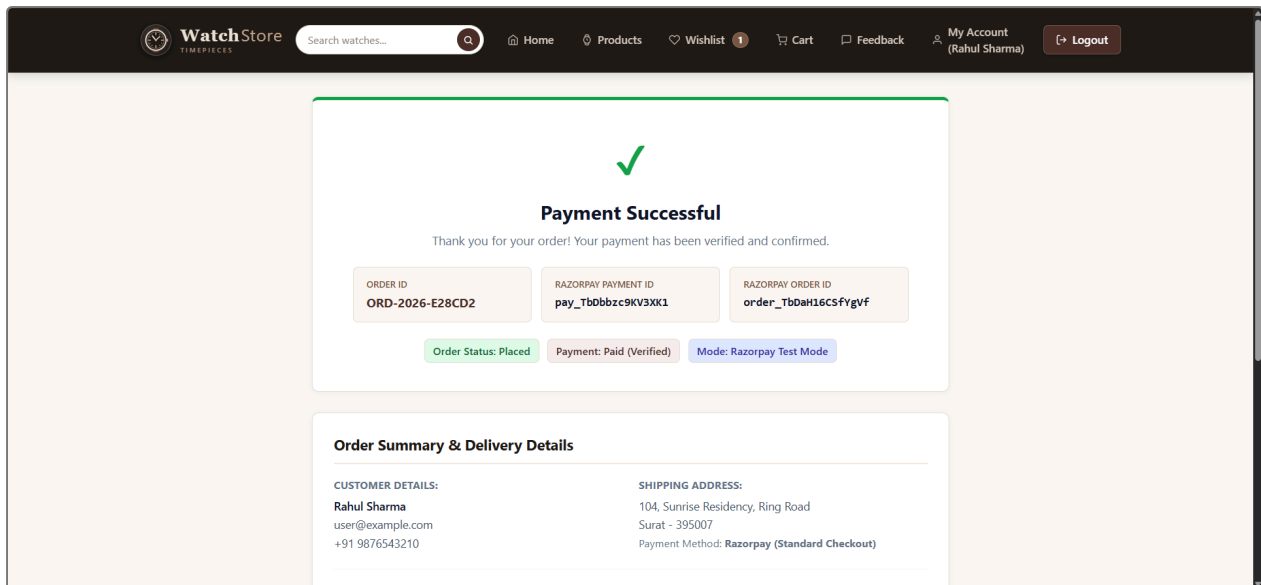


Figure 5.14: Order Confirmation & Payment Receipt (payment-success.php)

5.2.7 Customer Dashboard

Provides registered users with an overview of their account details, profile summary, and quick links to cart and orders.

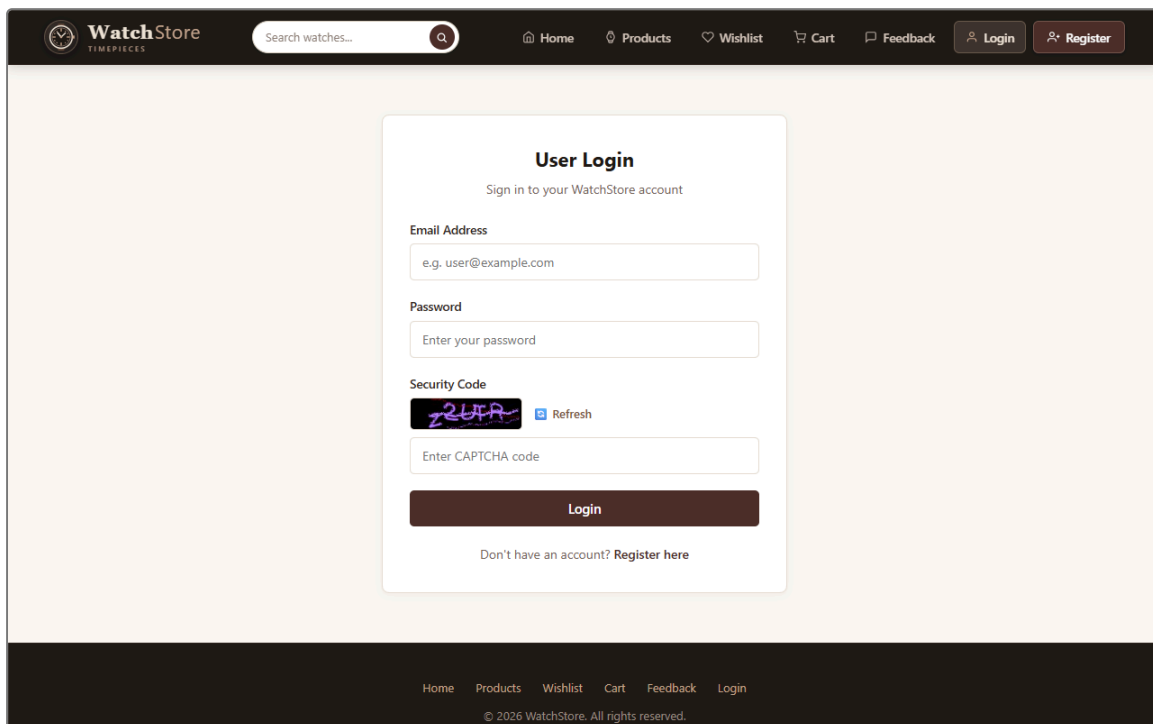


Figure 5.15: Customer Dashboard (dashboard.php)

5.2.8 Admin Dashboard

Displays key business metrics to administrators including total products, total brands, customer reviews, and quick links to manage catalog items.

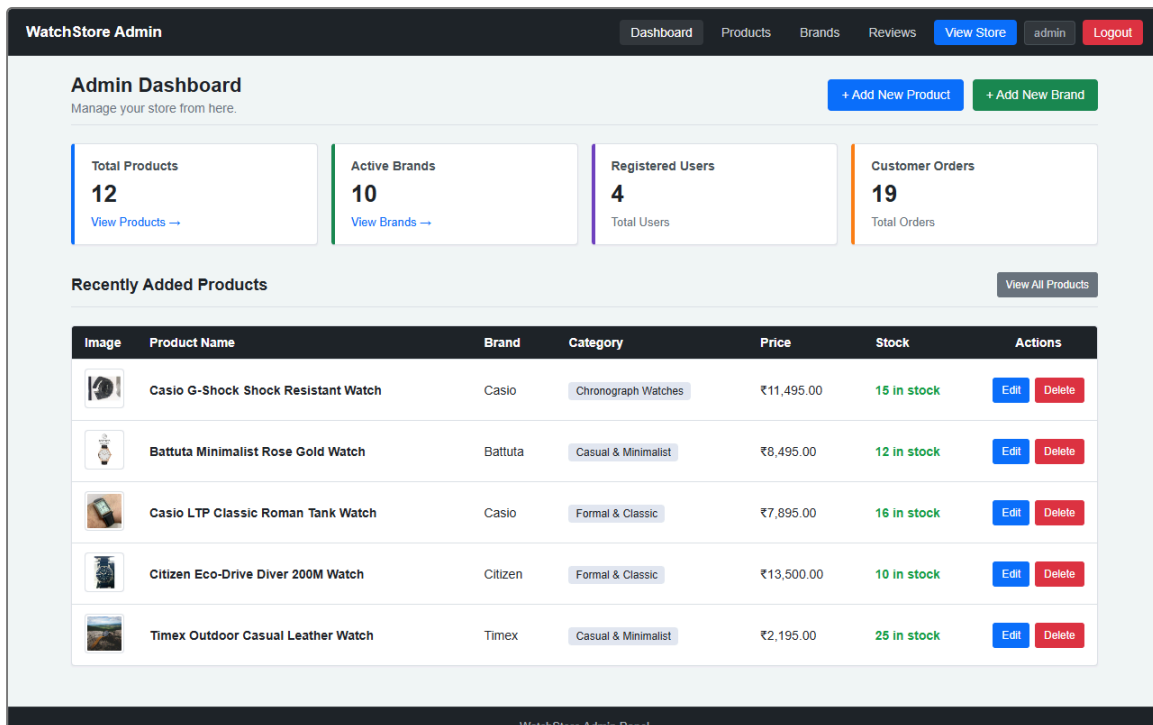


Figure 5.16: Admin Dashboard (admin/dashboard.php)

5.2.9 Admin Products Management Table

Displays all watches in a tabular format with options to edit details or delete products from the catalog.

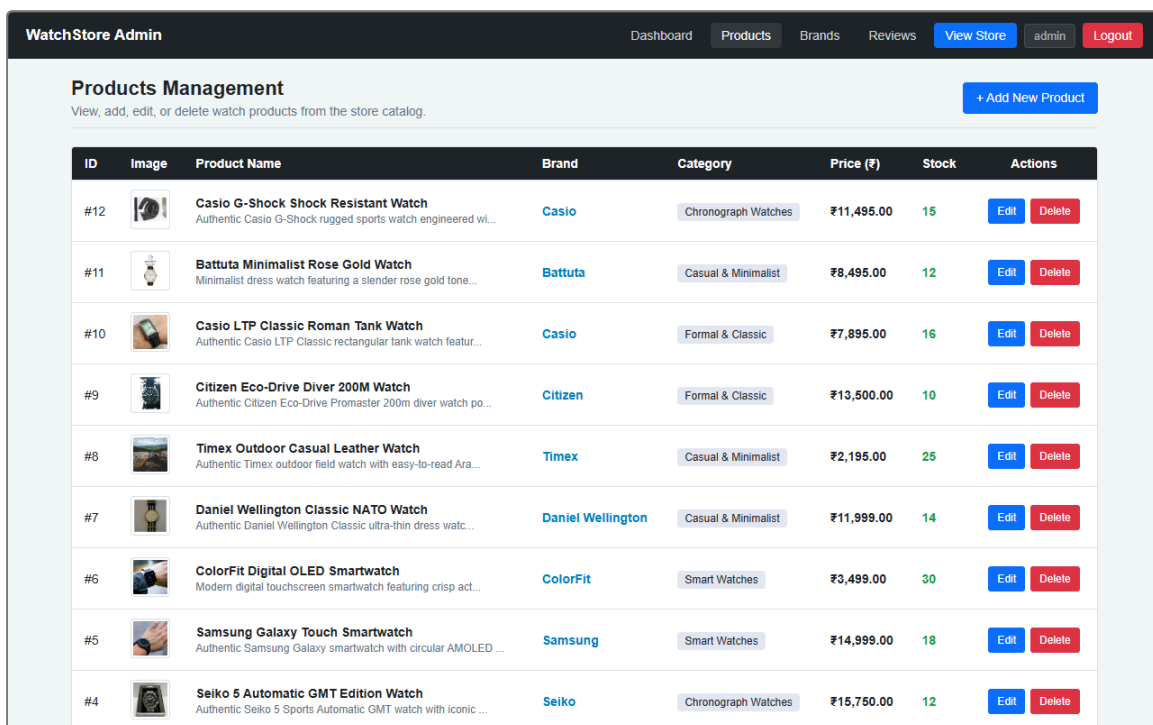


Figure 5.17: Admin Products Management (admin/products.php)

5.2.10 Admin Reviews Moderation Table

Displays customer reviews and star ratings with the capability to delete inappropriate or abusive feedback.

WatchStore Admin

DashboardProductsBrandsReviewsView StoreadminLogout

Customer Reviews

View and manage customer reviews and star ratings for watches.

ID	Product	User	Rating	Feedback / Comment	Date	Action
#3	 Titan Contemporary Quartz Watch	PRY pry@gmail.com	★★★★★ (5/5)	"good"	12 Sep 2026, 10:56 PM	Delete
#2	 Fossil Two-Tone Chronograph Watch	Saumya saumya@gmail.com	★★★★☆ (4/5)	"Looks good and works properly."	12 Sep 2026, 10:51 PM	Delete
#1	 Titan Contemporary Quartz Watch	Rahul Sharma user@example.com	★★★★★ (5/5)	"Very good watch and quality is nice."	12 Sep 2026, 10:51 PM	Delete

WatchStore Admin Panel

Figure 5.18: Admin Customer Reviews Moderation (admin/reviews.php)

6 SOFTWARE TESTING

6.1 Introduction

Software testing is the process of executing a program with the intent of finding errors, verifying that all system requirements are satisfied, and ensuring that user inputs produce expected outcomes under various conditions. For the Online Watches Website, testing was carried out at multiple levels to ensure reliability, data consistency, and proper integration with the database and payment gateway.

6.2 Objectives of Testing

- To verify that all customer registration and login validations work as intended.
- To ensure that product prices, item quantities, and order subtotals are calculated accurately without rounding errors.
- To verify that sessions prevent unauthorized access to customer dashboards and admin pages.
- To ensure that Razorpay test payments correctly create pending orders, verify cryptographic signatures, update order statuses, and decrement stock.
- To verify that product reviews and website feedback are stored properly in the database.

6.3 Levels of Testing

The testing process followed standard software engineering methodology:

- **Unit Testing:** Tested individual PHP functions, database query scripts, and form input validators in isolation (e.g. email regex validation, password hash generation).
- **Integration Testing:** Tested the interaction between interconnected modules, such as adding a product to the cart, transferring session cart data to the checkout controller, creating the order in the database, and updating stock quantities upon payment verification.
- **System Testing:** Tested the complete end-to-end user journey across the entire application, from landing on the home page as an anonymous visitor to registering, browsing, purchasing via Razorpay test mode, and viewing the order confirmation receipt.
- **Acceptance Testing:** Verified that the system meets all the functional objectives defined for the BCA Semester 5 WFS project.

6.4 Test Cases

The table below presents the actual test cases executed on the live WatchStore application along with their inputs, expected outcomes, actual results, and final pass/fail status.

Test ID	Test Case Description	Input / Action	Expected Result	Actual Result	Status
TC-01	User Registration with valid details	Name: Test User, Email: newuser@test.com, Mobile: 9876543210, Pass: pass123	Account created, redirected to login with success message	Account created successfully and redirected	PASS
TC-02	User Registration with existing email	Email: user@example.com	Show error message "Email already registered"	Error displayed, form not	PASS

Test ID	Test Case Description	Input / Action	Expected Result	Actual Result	Status
		(already in database)		submitted	
TC-03	User Login with correct credentials	Email: user@example.com, Pass: user123	Successful login, session created, redirect to dashboard	Logged in and redirected to dashboard	PASS
TC-04	User Login with incorrect password	Email: user@example.com, Pass: wrongpass	Show error message "Invalid email or password"	Error message displayed properly	PASS
TC-05	Browse Product Catalog	Navigate to products.php	Display all watch cards with name, brand, price, and image	All watch cards displayed properly	PASS
TC-06	Filter Products by Brand	Click "Titan" brand link (brand.php?name=Titan)	Display only Titan brand watches	Only Titan watches displayed	PASS
TC-07	View Watch Product Details	Click watch card on products.php (id=1)	Open product-details.php with full specs, price, and reviews	Product details page rendered correctly	PASS
TC-08	Add Watch to Wishlist	Click Wishlist button on product card	Item added to wishlist table for logged-in user	Item saved to wishlist and badge updated	PASS
TC-09	Add to Cart & Quantity Update	Click "Add to Cart" and change quantity to 2	Cart total updated to price * 2	Cart subtotal calculated accurately	PASS
TC-10	Checkout Form Validation	Submit checkout form with empty shipping address	Prevent submission and show "Please fill in all shipping details"	Validation triggered and error shown	PASS
TC-11	Razorpay Test Mode Order Creation	Submit valid checkout form	Server creates Razorpay order via cURL and launches checkout popup	Razorpay popup launched with correct amount	PASS
TC-12	Payment Signature Verification	Complete payment in Razorpay test modal	verify-payment.php validates HMAC SHA256 signature and updates order status	Signature verified, order status updated to Placed	PASS
TC-13	Display Order Success Receipt	Redirect to payment-success.php	Display order reference, payment ID, and ordered watch list	Receipt rendered with complete details	PASS
TC-14	Submit Product Review & Rating	Select 5 stars and enter comment on product-details.php	Review inserted into reviews table and displayed on page	Review saved and rendered immediately	PASS
TC-15	Submit Website Feedback	Fill feedback form with rating and message	Feedback record saved into feedback table	Feedback saved with success message	PASS
TC-16	Admin Login & Dashboard Access	Enter admin / admin123 on admin/login.php	Access granted to admin dashboard and management	Admin dashboard loaded	PASS

Test ID	Test Case Description	Input / Action	Expected Result	Actual Result	Status
			menu	successfully	
TC-17	Admin Add New Watch Product	Fill product add form and upload image	New watch added to products table and visible in store catalog	Product saved and listed in catalog	PASS

7 LIMITATION AND SYSTEM ENHANCEMENT

7.1 Limitation

While the Online Watches Website successfully implements the core features of an e-commerce platform, there are several limitations inherent to a student-level project:

- **Payment Gateway in Test Mode:** The Razorpay integration is configured in Test Mode using sandbox credentials. Real monetary transactions cannot be processed.
- **Local Server Hosting:** The system is currently hosted locally using XAMPP on localhost. It is not deployed on a public web server with a custom domain name.
- **Single Currency:** All watch prices and transactions are strictly denominated in Indian Rupees (INR ₹). Multi-currency conversion is not supported.
- **No Automated SMS Gateway:** Order confirmations and OTP verifications are handled through email and screen prompts; SMS notifications are not integrated due to paid API constraints.
- **Basic Inventory Management:** Stock count decreases when an order is placed, but automated supplier re-ordering or low-stock email alerts are not implemented.

7.2 System Enhancement

The following enhancements can be implemented in future iterations to transform this student prototype into a commercial-grade e-commerce system:

- **Live Payment Gateway Integration:** Transitioning from Razorpay Test Mode to Live Mode with production merchant API keys and automated webhook handlers.
- **Cloud Deployment with SSL:** Hosting the website on a cloud hosting provider (such as AWS, DigitalOcean, or Hostinger) with an SSL/TLS certificate (HTTPS) for secure data transmission.
- **SMS & WhatsApp Notifications:** Integrating SMS gateway APIs (such as Twilio or Fast2SMS) to send instant order confirmations, tracking links, and delivery updates directly to customers' mobile phones.
- **Discount Coupons & Promotional Offers:** Adding a promotional code module where administrators can create percentage or flat discount vouchers for festive sales.
- **Multi-Image Gallery & 360-Degree View:** Enhancing the watch details page with multiple high-resolution photos, zoom capabilities, and 360-degree interactive product views.
- **Automated Order Invoice PDF Generation:** Providing downloadable PDF tax invoices for customers directly from their dashboard.

8 BIBLIOGRAPHY

The following official documentation, textbooks, and online resources were referenced during the design, development, and documentation of the Online Watches Website project:

Books & Academic References:

1. Luke Welling and Laura Thomson, *PHP and MySQL Web Development*, 5th Edition, Addison-Wesley Professional, 2016.
2. Robin Nixon, *Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5*, 5th Edition, O'Reilly Media, 2018.
3. Jon Duckett, *HTML and CSS: Design and Build Websites*, John Wiley & Sons, 2011.
4. Abraham Silberschatz, Henry F. Korth, and S. Sudarshan, *Database System Concepts*, 7th Edition, McGraw-Hill, 2019.

Official Documentation & Online Resources:

1. **PHP Official Documentation:** The official PHP manual for syntax, session handling, password hashing, and MySQLi functions.
URL: <https://www.php.net/docs.php>
2. **MySQL Reference Manual:** Documentation for database schema design, SQL syntax, foreign key constraints, and indexing.
URL: <https://dev.mysql.com/doc/>
3. **Razorpay Payment Gateway Documentation:** Official integration guide for Razorpay Standard Checkout and Server-Side Order creation via cURL APIs.
URL: <https://razorpay.com/docs/payments/payment-gateway/web-integration/standard/>
4. **Mozilla Developer Network (MDN Web Docs):** Comprehensive reference for HTML5 semantic elements, CSS3 flexbox and grid layouts, and JavaScript APIs.
URL: <https://developer.mozilla.org/>
5. **W3Schools Web Tutorials:** Practical examples and reference materials for PHP, MySQL, HTML, and CSS.
URL: <https://www.w3schools.com/>